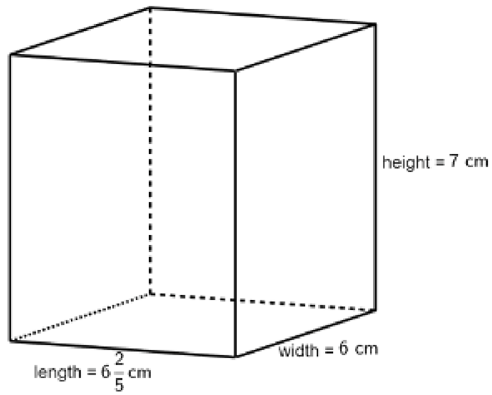
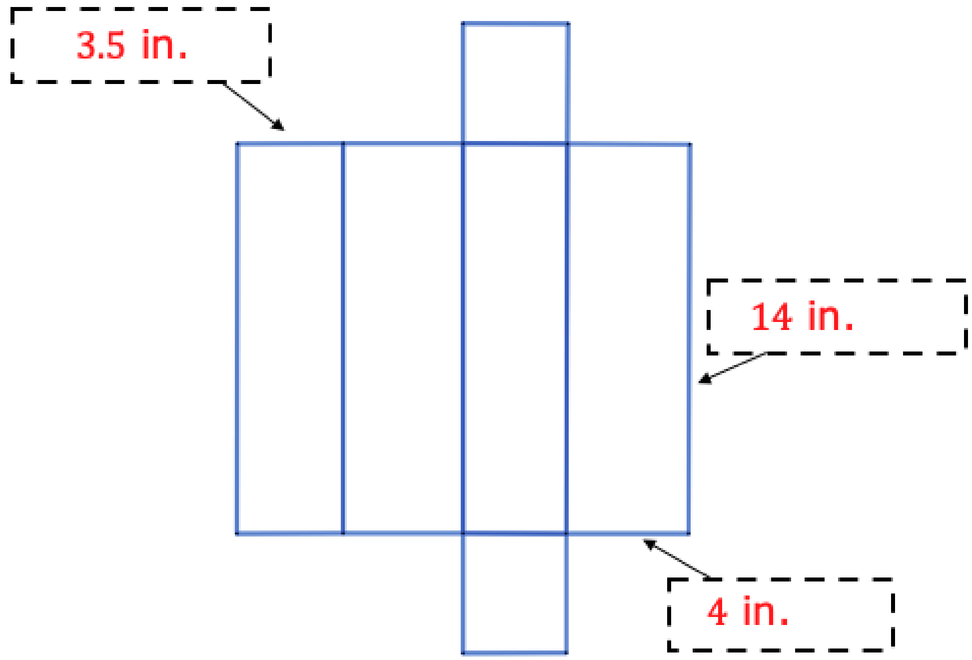
Grade 6 Accelerated Unit 8 Assessment Guide

General Information	The variety of questions that are possible for the benchmarks that are assessed in Unit 8 could not be covered in their entirety. To avoid a lengthy assessment, the number of real-world questions were limited. The assessment includes a question where students can show their conceptual understanding of packing cubes into a rectangular prism as well as their understanding of when to determine the volume or surface area of a figure in context. Teachers can use students' results on the Unit 7 assessment to determine and address potential gaps with finding area prior to concluding Unit 8. This assessment can then be used to determine if students have made progress in this area. Students may need guidance in how to show their work when they have a calculator. Giving students an example such as the one shown may help students organize their work while also helping in the grading process. Equipping students to self-scaffold a problem through their work is a valuable outcome to requiring students to show their work. work area of triangle: $\frac{1}{2}(16.3)(22.4) = 182.56$ area of triangle: $\frac{1}{2}(12.6)(30) = 189$ area of rectangle: $(21.5)(16.3) = 350.45$ Total: $182.56 + 189 + 350.45 = 722.01$ Teachers may wonder why the units are given in the answer boxes. This is done so that a student cannot claim that they used a different unit to find the answer. There is not benchmark or MTR that specifies that a student must show knowledge of the correct use of a unit. Although this is an important distinction students should make, it is not assessed on this assessment. If teachers prefer to incorporate this into their assessment, then it is suggested that the teacher makes the necessary edits to the questions, the answer boxes, and the point distributions. Teachers are cautioned from over-emphasizing units when distributing points.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1a	Benchmark(s)	MA.6.GR.2.3
4.25 inches		**Recommended Scoring** **(10 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**1b**	**Benchmark(s)**	MA.6.GR.2.3; MA.6.GR.2.4
The amount of cardboard used to construct the shipping box can be determined by finding the [○ volume / ○ surface area] of the box, while the amount of paper the shipping box can hold can be determined by finding the [○ volume / ○ surface area] of the box.		**Recommended Scoring** **(4 Total Points)**	Consider no partial credit.
Question	**2**	**Benchmark(s)**	MA.6.GR.2.4
340,928 square millimeters		**Recommended Scoring** **(20 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider breaking the points up as follows: • area of rectangle • area of left or right triangle • area of top or bottom triangle • total Surface Area

Question	3a	Benchmark(s)	MA.6.GR.2.3
A rectangular prism with dimensions in centimeters (cm) is shown.  a. Complete the statements. To find the volume of the prism, cubes with side lengths of ○ $\frac{1}{2}$ cm ○ $\frac{1}{5}$ cm ○ $\frac{1}{6}$ cm ○ $\frac{1}{7}$ cm can be used so that the length will have 32 cubes, the width will have 30 cubes, and the height will have 35 cubes. This results in 33,600 cubes, each with a volume of ○ $\frac{1}{8}$ ○ $\frac{1}{125}$ ○ $\frac{1}{216}$ ○ $\frac{1}{343}$ cubic centimeters.		**Recommended Scoring** **(24 Total Points)**	Consider giving students 4 points for each correct answer choice.
Question	**3b**	**Benchmark(s)**	MA.6.GR.2.3
$268\frac{4}{5}$ or 268.8 cubic centimeters		**Recommended Scoring** **(10 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Students may use cubes or the formula to determine the volume.

Question	4a	Benchmark(s)	MA.6.GR.2.4
		**Recommended Scoring** **(12 Total Points)**	Consider 4 points for each label. Consider deducting 1 point for each occurrence of not providing the units.

Question	4b	Benchmark(s)	MA.6.GR.2.4
Surface Area: 238 square inches		**Recommended Scoring** **(20 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider breaking the points up as follows: • area of top/bottom rectangle • area of left/right rectangle • area of front/back rectangle • total Surface Area